CONTROL ENGINEERING

1. Consider the model of a simple supply chain. A company C purchases, every month, a quantity of raw material $u(k)$. A fraction δ_1 of the raw material is discarded, and a fraction α_1 is shipped to a producer P. A fraction α_2 of product is sold by the producer P to a retailer R and a fraction δ_2 is discarded. The retailer R returns a fraction β_3 of defective products to the producer P every month, and sells a fraction γ_3 to customers.

   The model of the supply chain can described by the equations

$$\begin{aligned} x_1(k+1) &= (1-\alpha_1-\delta_1)x_1(k)+u(k), \\ x_2(k+1) &= \alpha_1 x_1(k)+(1-\alpha_2-\delta_2)x_2(k)+\beta_3 x_3(k), \\ x_3(k+1) &= \alpha_2 x_2(k)+(1-\beta_3-\gamma_3)x_3(k), \\ y(k) &= \gamma_3 x_3(k), \end{aligned}$$

   where k is time, in months, $x_1(k)$ is the raw material in C, $x_2(k)$ is the product in P, $x_3(k)$ is the product in R, and $y(k)$ is the product sold to customers.

   Note that the parameters δ_1, α_1, δ_2, α_2, β_3 and γ_3 are non-negative, not greater than one and are such that

$$0 \le 1-\alpha_1-\delta_1 \le 1, \qquad 0 \le 1-\alpha_2-\delta_2 \le 1, \qquad 0 \le 1-\beta_3-\gamma_3 \le 1.$$

   a) Write the matrices A, B, C and D of the underlying discrete-time system, with input u, states x_1, x_2, x_3, and output y, describing the supply chain. [4 marks]

   b) Compute the reachability matrix of the system and determine for which values of the parameters the system is reachable. [4 marks]

   c) Suppose $\beta_3 = 0$. Study the stability properties of the system, that is determine for which value of the parameters the system is asymptotically stable, stable, or unstable. Note that not all cases may be applicable: in particular explain why the system cannot be unstable.
   (Hint: do not compute the characteristic polynomial of the matrix A.)
   [6 marks]

   d) Suppose $\alpha_1 = 0$. Determine the set of all states that can be reached from zero. Determine, if possible, conditions under which the system is controllable.
   [6 marks]

2. Consider the linear, time-invariant, single-input, continuous-time, system described by the equations

$$\dot{x} = Ax + Bu,$$

with state $x(t) \in \mathbb{R}^2$, input $u(t) \in \mathbb{R}$, and

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Suppose one would like to design a state feedback control law, described by the equation $u = Kx$ such that the closed-loop characteristic polynomial is $p_d(\lambda) = \lambda^2 + a_{d1}\lambda + a_{d2}$. Let $p(\lambda) = \lambda^2 + a_1\lambda + a_2$ be the characteristic polynomial of the matrix A.

The state feedback gain K can be designed using two formulae: the Bass-Gura formula and the Ackermann formula. The Bass-Gura formula is given by the equation

$$K_{bg} = -\begin{bmatrix} a_{d1} - a_1 & a_{d2} - a_2 \end{bmatrix} \Omega^{-1}\mathscr{R}^{-1},$$

where

$$\Omega = \begin{bmatrix} 1 & a_1 \\ 0 & 1 \end{bmatrix}$$

and $\mathscr{R}$ is the reachability matrix of the system. The Ackermann formula is given by the equation

$$K_a = -\begin{bmatrix} 0 & 1 \end{bmatrix} \mathscr{R}^{-1} p_d(A).$$

a) Show that the considered system is controllable and compute the reachability matrix. [2 marks]

b) Consider the problem of determining the state feedback gain matrix K such that all closed-loop eigenvalues are placed at -1.

i) Compute the desired closed-loop characteristic polynomial $p_d(\lambda)$. [2 marks]

ii) Compute the state feedback gain matrix K using Bass-Gura formula and verify that indeed the resulting K assigns the closed-loop eigenvalues at the desired location. [4 marks]

iii) Compute the state feedback gain matrix K using Ackermann formula. Explain why the resulting gain matrix coincides with that computed in part b.ii). [4 marks]

iv) Suppose now that in the computation of the inverse of the reachability matrix $\mathscr{R}$ one commits some numerical error, that is the computed inverse of the reachability matrix is given by

$$\hat{\mathscr{R}}^{-1} = \mathscr{R}^{-1} + \varepsilon I,$$

where I is the identity matrix and ε is a constant scalar parameter. Use $\hat{\mathscr{R}}^{-1}$ instead of $\mathscr{R}^{-1}$ to recompute the state feedback gain matrix using both the Bass-Gura formula and the Ackermann formula. Let $\hat{K}_{bg}$ and $\hat{K}_a$ be the resulting gain matrices. Compute the resulting closed-loop matrices, that is the matrices $A + B\hat{K}_{bg}$ and $A + B\hat{K}_a$.

Determine conditions on ε such that the eignevalues of $A + B\hat{K}_{bg}$ are in the left-half of the complex plane. Similarly, determine conditions on ε such that the eigenvalues of $A + B\hat{K}_a$ are in the left-half of the complex plane. Discuss which of the two given formulae, if any, is more robust against errors in the computation of the inverse of the reachability matrix.

[8 marks]

3. Consider the linear, time-invariant, single-input, discrete-time, system described by the equations

$$x^+ = Ax + Bu, \qquad y = Cx,$$

with state $x(t) \in \mathbb{R}^2$, input $u(t) \in \mathbb{R}$, output $y(t) \in \mathbb{R}$, and

$$A = \begin{bmatrix} 0 & 1 \\ a_{21} & a_{22} \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad C = \begin{bmatrix} c_1 & c_2 \end{bmatrix}.$$

The parameters a_{21}, a_{22}, c_1 and c_2 are constant and unknown. To compute these unknown parameters one could run an experiment (this makes sense if the input and output signals are precisely measured, if the initial condition is exactly known, and if the underlying system is reachable and observable). To this end, suppose that one knows that the initial state of the system is

$$x(0) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

and that applying the input sequence

$$u(0) = 1, \qquad u(1) = -1, \qquad u(2) = 0,$$

yields the output sequence

$$y(0) = 1, \qquad y(1) = -1, \qquad y(2) = -3/2, \qquad y(3) = 3.$$

a) Study the reachability and observability properties of the system as a function of the parameters a_{21}, a_{22}, c_1 and c_2 . [2 marks]

b) Compute $y(0)$, $y(1)$, $y(2)$, and $y(3)$ as a function of the parameters a_{21}, a_{22}, c_1 and c_2 and of the given input sequence and initial condition. [6 marks]

c) Using the information provided on the values of the measured output sequence determine the parameters a_{21}, a_{22}, c_1 and c_2 which are consistent with such information.
(Hint: you need to write four equations to be solved for c_2, $c_1 + a_{22}$, $a_{21} - 2a_{22}$, and a_{22}.)
Show that the parameters are such that the system is observable.
[6 marks]

d) Suppose, instead, that for the same input sequence and the same initial condition given above the resulting measured output sequence is

$$y(0) = 1, \qquad y(1) = 0, \qquad y(2) = -1, \qquad y(3) = 1.$$

Show that in this case it is not possible to uniquely determine the coefficients a_{21}, a_{22}, c_1 and c_2 and explain why this is the case. [6 marks]

4. Consider a linear, single-input single-output, continuous-time, time-invariant, system described by the equations

$$\dot{x} = Ax + Bu, \qquad y = Cx,$$

with state $x(t) \in \mathbb{R}^n$, input $u(t) \in \mathbb{R}$, and output $y(t) \in \mathbb{R}$.

Suppose one wishes to control the system with a mixture of state feedback and integral action on the ouput tracking error, that is the error between the reference signal r, with $r(t) \in \mathbb{R}$, and the output signal y.

This can be achieved setting

$$u = K_{sf}x + K_I v,$$

where K_{sf} is the state feedback gain, K_I is the integral gain, and the signal v is such that

$$\dot{v} = r - y.$$

a) Write the equations of the interconnected system with state x and v, input r and output y. [2 marks]

b) Show that the "A" matrix A_{int} of the interconnected system can be written as

$$A_{int} = A_{aug} + B_{aug}K_{aug},$$

where $K_{aug} = \begin{bmatrix} K_{sf} & K_I \end{bmatrix}$. Determine explicitely the matrices A_{aug} and B_{aug}. [4 marks]

c) To arbitrarily assign the eigenvalues of the matrix A_{int} one has to check that the pair $\{A_{aug}, B_{aug}\}$ is reachable. To study reachability of the pair $\{A_{aug}, B_{aug}\}$ one could proceed as follows.

i) Write the reachability pencil associated to the pair $\{A_{aug}, B_{aug}\}$. [2 marks]

ii) Show that the reachability pencil associated to the pair $\{A_{aug}, B_{aug}\}$ has full rank for all non-zero eigenvalues of the matrix A provided the pair $\{A, B\}$ is reachable. [4 marks]

iii) Show that matrix A_{aug} has at least one eigenvalue at zero. [2 marks]

iv) Show that the reachability pencil associated to the pair $\{A_{aug}, B_{aug}\}$ has full rank for all zero eigenvalues of the matrix A_{aug} provided the matrix

$$\begin{bmatrix} -A & B \\ C & 0 \end{bmatrix}$$

has rank $n+1$. [4 marks]

v) Hence argue that the pair $\{A_{aug}, B_{aug}\}$ is reachable provided the pair $\{A, B\}$ is reachable and the rank condition in part c.iv) holds. [2 marks]